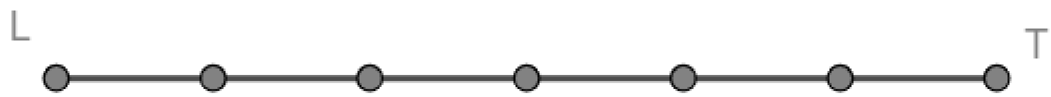
Geometry
Unit 13
Lesson 4 Practice

Name____________________
Date__________________
Period__________

Use the following information for questions 1-4. Directed line segment LT is shown.



1. How many equal parts is $\overline{LT}$ partitioned into?

2. Make a mark where $\overline{LT}$ is partitioned into a $1:3$ ratio. Call this point K.

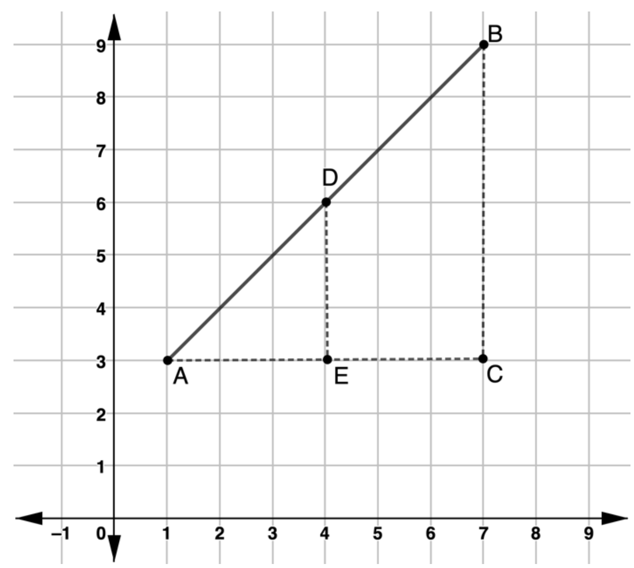
3. Complete the statement.

   Point K is _________ of the distance from L to T.

4. Complete the equation that can be used to find the distance of $\overline{LT}$.

$$LK = \_\_\_\_ LT$$

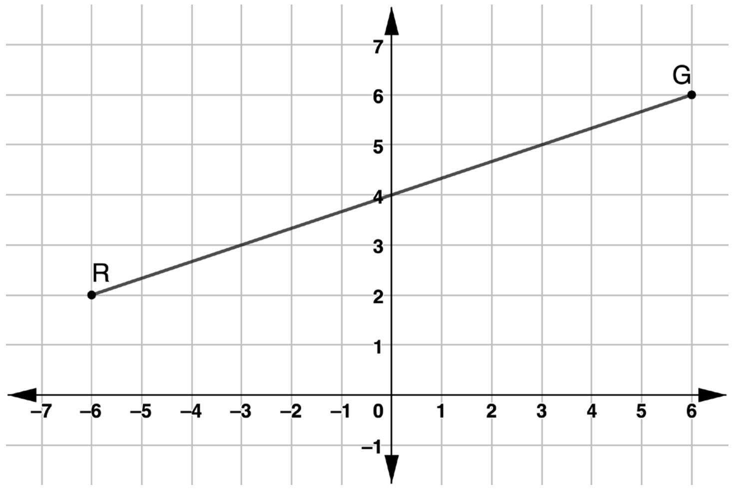
5. Point D partitions directed line segment AB in a $3:3$ ratio. Points $A(1,3)$, $E(4,3)$, $D(4,6)$, $B(7,9)$, and $C(7,3)$ are shown on the coordinate grid.



Use the distances to write the numeric ratio for each segment ratio. Then, choose the type of ratio the segment ratio represent.

Segment Ratio	**Numeric Ratio**	**Type of Ratio**
$AE:EC$		○ part: part ○ part: whole
$AE:AC$		○ part: part ○ part: whole
$DE:BC$		○ part: part ○ part: whole
$AD:DB$		○ part: part ○ part: whole

Use the following information for questions 6-8. Directed line segment RG is shown with endpoints at point $R(-6,2)$ and point $G(6,6)$.



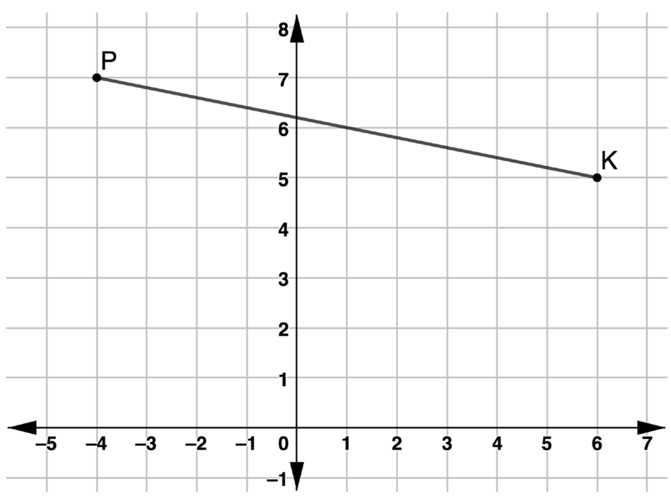
Directed line segment RG is partitioned $\frac{3}{4}$ of the distance from R to G at point T.

6. What is the horizontal distance of $\overline{RT}$?

7. What is the vertical distance of $\overline{RT}$?

8. What are the coordinates of point T?

Use the following information for questions 9-10. Directed line segment PK has endpoints at $P(-4,7)$ and $K(6,5)$.



Directed line segment PK is partitioned $\frac{2}{5}$ of the distance from P to K at point R.

9. What is the horizontal and vertical distance of $\overline{PR}$?

10. What are the coordinates of point R?